

Paper Reading Notes

Building Services and Information

Building Science / Building Information
Screening and Literature Library

Paper ID: permanent library identifier (P001, P002, ...).

BibTeX key: bibliography identifier.

Citation number: assigned in citation order; independent of Paper ID.

目次

1	A. Building Environment and IEQ	2
	P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks	2
	P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space	2
2	B. HVAC and Building Services	3
3	C. Building Energy and ZEB	4
4	D. BMS, BAS, BACS and Building Automation	5
5	E. BIM and Semantic Information	6
6	F. Digital Twin	7
7	G. Sensors, IoT and Data Integration	8
8	H. Fault Detection, Diagnostics and Maintenance	9
9	I. AI and Machine Learning for Buildings	10
10	J. Retrieval, Query, LLM and Agent	11
11	K. Occupancy and Human-Building Interaction	12
12	L. Reviews, Frameworks and Standards	13
13	Uncategorized	14

1 A. Building Environment and IEQ

P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks

Authors: Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, Eric Schuldenfrei

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Spatial-temporal prediction, CO2 concentration, Education facility, Space reconfigurability, BIM, Graph neural networks, Multi-modal deep learning

■**主な主張** BIM 由来の時変レイアウトと Voronoi ベースの CO2 センサ隣接グラフを multi-modal deep learning で統合し, 2 時間先予測で MAE 11.25 ppm を達成し, 比較対象中で次点の TGCN の 15.10 ppm より約 25% 低い MAE を示した. [1]

■**新規性** 静的レイアウトを前提とする既往の GNN 型 CO2 予測に対し, BIM から時変の間仕切り構成をテンソル化し, センサグラフ文脈と空間構成文脈を二枝の MMDL で統合した点に新規性がある.

■**位置付け** 教育施設の IAQ 予測を主目的としつつ, BIM 幾何情報を予測モデルの入力文脈として利用するため, Building Environment and IEQ を主分類, AI and Machine Learning for Buildings および BIM and Semantic Information を副分類として位置付ける.

P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space

Authors: Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, Jack C.P. Cheng

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

Keywords: Graph neural network, Spatial-temporal prediction, Spatial-temporal feature interactions, Indoor Air Quality, Indoor occupant, Dynamic graph learning, Occupant-in-loop

■**主な主張** 提案 AST-GCGRU は, 6 h および 12 h の multi-step prediction で graph-level R2 0.898 および 0.812 を示し, GCGRU の 0.860 および 0.699 を上回った. 12 h では occupant node R2 0.59, CO2 node R2 0.73 で, GCGRU の 0.32, 0.58 より高かった. [2]

■**新規性** 距離ベースの固定 graph ではなく, partial correlation で初期化した graph を ST-Att で学習中に更新し, time-lag nodes により 30 min CO2 sensor と 6 min occupant sensor の異なる時間解像度を保持したまま, CO2 と occupancy を同一 graph で同時予測・interaction analysis する構成を提示した.

■**位置付け** 大空間における IAQ と occupancy の同時予測・相互作用分析を主目的とするため, Building Environment and IEQ を主分類とし, AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration を副分類として位置付ける.

2 B. HVAC and Building Services

3 C. Building Energy and ZEB

4 D. BMS, BAS, BACS and Building Automation

5 E. BIM and Semantic Information

6 F. Digital Twin

7 G. Sensors, IoT and Data Integration

8 H. Fault Detection, Diagnostics and Maintenance

9 I. AI and Machine Learning for Buildings

10 J. Retrieval, Query, LLM and Agent

11 K. Occupancy and Human-Building Interaction

12 L. Reviews, Frameworks and Standards

13 Uncategorized

References

- [1] Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, and Eric Schuldenfrei. Spatial-temporal co2 concentration prediction in education facilities with dynamic layouts using bim and multi-modal graph neural networks. *Building and Environment*, 297:114528, 2026.
- [2] Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, and Jack C.P. Cheng. Graph-based spatial-temporal prediction and feature interaction analysis of co2 and occupant in large indoor space. *Building and Environment*, 280:112963, 2025.